14.H2 Interpreting Relative Frequency Tables and Segmented Bar Graphs

Lesson Introduction

Benchmarks	MA.912.DP.3.2 Given marginal and conditional relative frequencies, construct a two-way relative frequency table summarizing categorical bivariate data. MA.912.DP.3.3 Given a two-way relative frequency table or segmented bar graph summarizing categorical bivariate data, interpret joint, marginal, and conditional relative frequencies in terms of a real-world context. MA.912.DP.1.3 Explain the difference between correlation and causation in the contexts of both numerical and categorical data.	Learning Targets	• I can construct a two-way frequency table. • I can interpret joint, marginal, and conditional relative frequencies in a real-world context.

Benchmark Coherence	Building On	Working Toward
	N/A	MA.912.DP.4.3 Calculate the conditional probability of two events, and interpret the result in terms of its context. MA.912.DP.4.4 Interpret the independence of two events using conditional probability.

Essential Questions	• How do you construct a two-way relative frequency table? • How do you interpret joint, marginal, and conditional relative frequencies?
Lesson Narrative	This lesson builds from Lesson H1 where students learned how to use provided data to construct a two-way relative frequency table. In this lesson, students will continue to construct two-way relative frequency tables along with learning how to use the two-way relative frequency table to interpret joint, marginal, and conditional relative frequencies.

Component Summary				
	14.H2.1 Warm-Up (~5 min.)	Notice and Wonder from a segmented bar graph. *(SE p. 331)*	14.H2.3 Try It (10-15 min.)	Interpret relative frequencies from a two-way relative frequency table. *(SE p. 334)*
	14.H2.2 Collaboration (20-25 min.)	Create and interpret data from a two-way relative frequency table. *(SE p. 332)*	14.H2.4 Wrap-Up (~5 min.)	Ticket Out the Door on interpreting relative frequencies. *(SE p. 335)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: • Conditional Relative Frequency • Association	• N/A	• (Optional) Equipment to display a two-way frequency table. • (Optional) Additional data that could show that association does not imply causation.

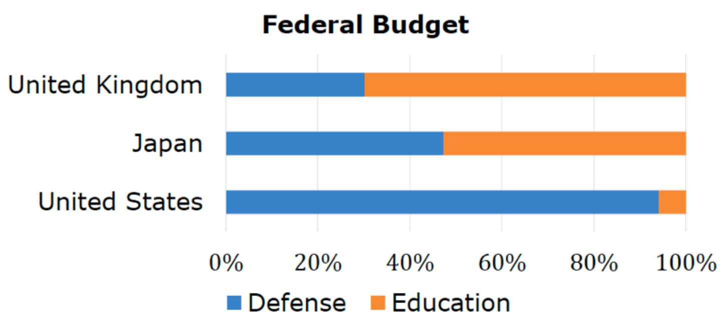
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may become stuck when using conditional relative frequencies to find joint relative frequencies when completing a two-way frequency table.
Students may not realize they need to find the conditional relative frequency by dividing the percent admitted by their marginal relative frequencies in 14.H2.3 Try It.</td><td>Consider working through this lesson to find sticking points for students when calculating and interpreting data.
Consider pulling other data that shows an association between variables, but would not necessarily mean that one caused the other.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
14.H2.3 Try It provides a good opportunity to use the Think-Pair-Share routine. During the share portion of this routine, encourage students to access the reasonableness of their answer by justifying their results by explaining their method and process.</td><td>MA.K12.MTR.2.1: Discussion and Reflection
14.H2.3 Try It positions students to “construct possible arguments based on evidence.” In this activity, students find and interpret relative frequencies to draw a conclusion based on the data.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson provides students the opportunity to interpret data from multiple representations. In 14.H2.3 Collaboration, students construct a two-way relative frequency using the provided data, but at the end of the activity students are provided the chance to analyze the similarities and differences in the two displays. It also provides students a chance to check their values they put in for their two-way relative frequency table.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

14.H2.1 Warm-Up: Notice and Wonder

Component Overview: *During this Notice and Wonder, students will practice analyzing data from a segmented bar graph.*

14.H2.1 Warm-Up: Notice and Wonder

The segmented bar graph represents the distribution of where some of the 2009 federal budgets were expected to go.



1. What do you notice?

   Answers will vary.

2. What do you wonder?

   Answers will vary.

14.H2.1 Warm-Up is intended to be done independently and to be easily accessible by any student. The students' responses may vary from the provided answer in the key. Consider asking students that typically do not like to share during class to share their responses to questions 1 and 2.

14.H2.2 Collaboration: Two-Way Relative Frequency Tables

Component Overview: *14.H2.2 Collaboration provides students with the opportunity to practice creating a two-way relative frequency table with a partner from data provided. It then goes deeper into this lesson's focus on interpreting the data from the two-way relative frequency table.*

14.H2.2 Collaboration: Two-Way Relative Frequency Tables

Data was collected from a 2019 random sample of traffic accidents in Chicago. Data on the road conditions at the time of the accidents (dry, wet/slushy, icy) and the amount of damage caused ($500 or less, $501–$1,500, more than $1,500) is summarized.

- 27.76% of the accidents involved damage of $500 or less.
- 28.77% of the accidents involved damage of $501 to $1,500.
- 78.43% of the accidents involved dry road conditions.
- 20.40% of the accidents involved wet/slushy road conditions.
- Of the accidents with damage of more than $1,500, 69.77% involved dry road conditions.
- Of the accidents with damage of $501–$1,500, 80.22% involved dry road conditions.
- 24.79% of the accidents with icy road conditions had damage of $501–$1,500.
- 10.26% of the accidents with icy road conditions had damage of $500 or less.

1. Use the data provided to construct a two-way relative frequency table. Round values to the nearest hundredth.

Traffic Accident Data

	Damage			
Road Conditions	**$500 or less**	**$501-$1,500**	**More than $1,500**	**Total**
Dry	25.02%	23.08%	30.33%	78.43%
Wet/Slushy	2.62%	5.40%	12.38%	20.40%
Icy	0.12%	0.29%	0.76%	1.17%
Total	27.76%	28.77%	43.47%	100%

Students may become stuck when completing the two-way frequency table from the given data. Consider having students label each bulleted data as either a joint, marginal, or conditional relative frequency to better help them decide how to complete the two-way frequency table.

Consider the following guiding questions to help students determine their next steps:

- *When completing a two-way relative frequency table, what is the total percentage for the marginal frequencies?*
- *During Lesson H1, how did you use conditional relative frequencies to find the joint relative frequencies?*

14.H2.2 Collaboration: Two-Way Relative Frequency Tables (continued)

2. Interpret the joint relative frequency for dry road conditions with damage of $500 or less.
The joint relative frequency for dry road conditions that result in damage of $500 or less is 25.02%. This means that 25.02% of the accidents involved dry road conditions and damage of $500 or less.

3. Interpret the marginal relative frequency for traffic accidents that had icy road conditions.
The marginal relative frequency for traffic accidents that had icy road conditions is 1.17%. This means that in 1.17% of the accidents, the road was icy.

4. When the road conditions are icy, is the likelihood of the different amounts of damage the same? Explain your reasoning using conditional relative frequencies.
Answers will vary. The likelihood of the different amounts of damage is not the same for icy road conditions. Of the accidents involving icy conditions, about 65% resulted in more than $1,500 of damage, about 25% resulted in $501–$1,500 of damage, and only about 10% resulted in damage of $500 or less. Accidents involving icy conditions resulted in more than $1,500 of damage 6 times more often than they resulted in damage of $500 or less.

5. Does there appear to be an association between road conditions and the amount of damage caused?
Answers will vary. There seems to be an association between road conditions and the amount of damage caused. Over half of the accidents that occurred in wet/slushy conditions or icy conditions resulted in more than $1,500 of damage. Only 39% of the accidents that occurred in dry conditions resulted in more than $1,500 of damage; however, in wet/slushy conditions, 61% of the accidents resulted in more than $1,500 of damage, and in icy conditions, 65% resulted in more than $1,500 of damage.

6. Does this mean that more hazardous road conditions cause more damage in traffic accidents?
Answers will vary. Not necessarily. Hazardous road conditions seem to play a role in the amount of damage caused in a traffic accident. However, there are other factors that can affect how much how much damage is caused in traffic accidents.

Take Turns
Have students take turns answering questions 2-6. Students should not just answer every other question but listen to each other's justification as to why they interpreted the data the way they did, and the other students can agree or disagree, providing justification as to why.

Facilitate classroom discussion following the Take Turns routine. Use discussion guides to frame conversations through students' personal experiences, differing perspectives, and their interpretation of the data in questions 2-6.

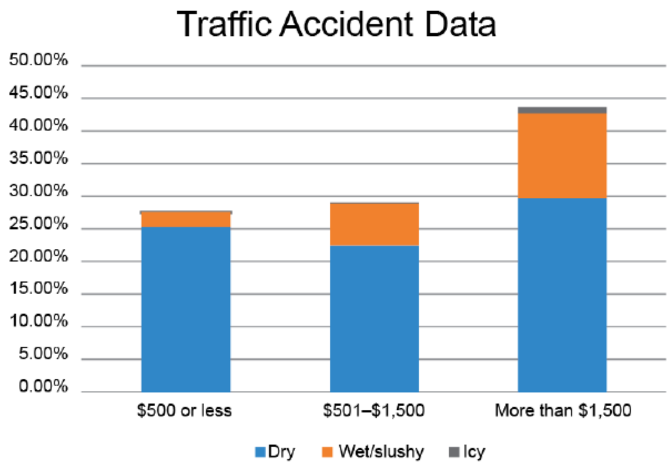
Have students revisit resources and Study Expert videos from Lesson 4 for additional support with terms like "association."

14.H2.2 Collaboration: Two-Way Relative Frequency Tables (continued)

Association does not imply causation!
If there is an association between two variables, it does not mean that one causes the other.
Two variables can be associated without having a causal relationship.

Another way to represent bivariate categorical data from a two-way relative frequency table is with a segmented bar graph. These graphs are helpful in visualizing comparisons between different variables by identifying how much larger or smaller different parts of each bar are compared to other bars.

The data from the two-way frequency table can be displayed in a segmented bar graph as shown.



7. With your partner, discuss the similarities and differences between the two-way relative frequency table and the segmented bar graph. Summarize your discussion.

Answers will vary. The two-way relative frequency table and the segmented bar graph show the same data. The segmented bar graph makes it easier to visually compare the relative frequencies.

Consider displaying the two-way relative frequency table for the whole class to see, so that students can easily discuss the similarities and differences without having to flip pages to look back at the table.

Consider the guiding questions to prompt students to make connections between the frequency table and the segmented bar graph.

- *Compare the frequency table to the segmented bar graph. What looks the same? What looks different?*
- *What do the similarities and differences make you think about which representation might be used in different scenarios? Why?*
- *For example, how does the segmented bar graph make you think differently about the data, comparatively to the frequency table? Why?*

14.H2.3 Try It: Interpreting Relative Frequencies

Component Overview: *Students will practice independently interpreting relative frequencies from a two-way relative frequency table.*

14.H2.3 Try It: Interpreting Relative Frequencies

The General Social Survey (GSS) interviews a random sample of Americans every two years. In one question, respondents are asked to rate their general happiness level. Sample results for 1972, 1998, and 2021 are summarized in the table.

	1972	1998	2021	Total
Very Happy	486	891	783	2,160
Pretty Happy	855	1,575	2,308	4,738
Not Too Happy	265	340	923	1,528
Total	1,606	2,806	4,014	8,426

Consider using a Think-Pair-Share routine for 14.H2.3 Try It. Have students first complete 14.H2.3 Try It independently, then pair students to discuss how they interpreted the data for questions 1 – 8, before bringing the class back together to share and discuss their interpretations.

1. After reviewing the two-way frequency table, Diana said, "It seems like people in 2021 were a lot happier than in 1972."

   a. Why do you think Diana said this?
   Answers will vary. The raw number of people who responded that they were very happy or pretty happy in 2021 was significantly higher than in 1972.

   b. Explain why this may not be an accurate conclusion to draw from the data.
   Answers will vary. The number of respondents in 2021 was significantly higher than in 1972, so looking at the raw numbers may not provide an accurate picture of the proportion of people in each category. A relative frequency table would be better to use in this case.

2. Use the data to create a two-way relative frequency table. Round values to the nearest tenth of a percent.

	1972	1998	2021	Total
Very Happy	5.8%	10.6%	9.3%	25.6%
Pretty Happy	10.1%	18.7%	27.4%	56.2%
Not Too Happy	3.1%	4.0%	11.0%	18.1%
Total	19.1%	33.3%	47.6%	100.0%

For questions 5-7, students may not realize they need to find the conditional relative frequency by dividing the percent admitted by their marginal relative frequencies.

14.H2.3 Try It: Interpreting Relative Frequencies (continued)

3. Interpret the value 9.3% in context of the data.
   Respondents who said they were very happy in 2021 make up 9.3% of the people surveyed in the 3 years represented.

4. Interpret the value 33.3% in context of the data.
   The people who responded in 1998 made up 33.3% of the total number of people surveyed in the 3 years represented.

5. Interpret the value 25.6% in context of the data.
   25.6% of the people surveyed over the 3 years shown reported they were very happy.

6. Interpret the value $\frac{27.4\%}{47.6\%}$ in context of the data.
   57.6% of the respondents from 2021 reported they were pretty happy.

7. What percentage of the respondents from 1972 reported they were very happy?
   30.3% of the respondents from 1972 reported they were very happy.

8. What percentage of the respondents from 1998 reported they were very happy?
   31.8% of the respondents from 1998 reported they were very happy.

9. What percentage of the respondents from 2021 reported they were very happy?
   19.5% of the respondents from 2021 reported they were very happy.

10. What can be concluded about the general happiness of Americans from the two-way relative frequency table?
    Answers will vary. In general, a similar percentage of Americans reported they were very happy in 1972 and 1998, but the percentage dropped significantly in 2021. The percentage of respondents each year who reported being pretty happy was similar, ranging from about $53\%–58\%$. The percentage of respondents who reported they were not too happy was the highest in 2021, at 23% of respondents. It is difficult to draw conclusions about the entire population from this small of a sample, but the survey data collected suggests Americans may have been less happy overall in 2021 than in 1972 or 1998.

14.H2.4 Wrap-Up: Ticket Out the Door

Component Overview: *In 14.H2.4 Wrap-Up, students will interpret a joint and conditional relative frequency about the Titanic.*

14.H2.4 Wrap-Up: Ticket Out the Door

Consider using this formative assessment with Lesson 3's Ticket Out the Door to determine if students show growth in interpreting data from a display.

1. The RMS Titanic was a British passenger liner operated by the White Star Line that sank in the North Atlantic Ocean on April 15, 1912, after striking an iceberg during her maiden voyage from Southampton to New York City.

   The two-way relative frequency table classifies the passengers as whether the person survived (Yes or No) and the class of the ticket they had (1^{st}, 2^{nd}, 3^{rd}, or Crew).

	Survived		
Ticket	**No**	**Yes**	**Total**
1st	5.5%	9.2%	14.8%
2nd	7.6%	5.4%	12.9%
3rd	24.0%	8.1%	32.1%
Crew	30.6%	9.6%	40.2%
Total	67.7%	32.3%	100.0%

   a. What does the value 30.6% mean in context of the data?
   30.6% is the joint relative frequency for crew who did not survive. 30.6% of all passengers were crew members who did not survive.

   b. What does the value $\frac{30.6\%}{40.2\%}$ mean in context of the data?
   This value means that 76.12% of all crew members on board the Titanic did not survive the wreck.